

# Homework 11: Longitudinal Data

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## Problem 27: AR(1) model (2 points)

### Deriving the log-likelihood

We are given a first-order autoregressive process AR(1) with the conditional distribution:

$$P(X^t = x^t \mid X^{t-1} = x^{t-1}) = \mathcal{N}(x^t \mid Ax^{t-1}, \Gamma),$$

where  $X^t \in \mathbb{R}^n$ ,  $A \in \mathbb{R}^{n \times n}$ , and  $\Gamma = I_n$  (the  $n \times n$  identity matrix).

The density of a multivariate normal  $\mathcal{N}(x \mid \mu, \Sigma)$  is:

$$p(x \mid \mu, \Sigma) = (2\pi)^{-n/2} |\Sigma|^{-1/2} \exp\left(-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu)\right).$$

Since  $\Gamma = I_n$ , we have  $|\Gamma| = 1$  and  $\Gamma^{-1} = I_n$ . Hence each conditional density simplifies to:

$$p(x^t \mid x^{t-1}, A) = (2\pi)^{-n/2} \exp\left(-\frac{1}{2}(x^t - Ax^{t-1})^\top (x^t - Ax^{t-1})\right).$$

By the Markov property of the AR(1) process, the joint likelihood of the observations  $x^1, \dots, x^T$  (conditioned on  $x^0$ ) factorizes as:

$$L(A) = \prod_{t=1}^T p(x^t \mid x^{t-1}, A).$$

Taking the logarithm:

$$\ell(A) = \log L(A) = \sum_{t=1}^T \log p(x^t \mid x^{t-1}, A).$$

Substituting the Gaussian density:

$$\ell(A) = \sum_{t=1}^T \left[ -\frac{n}{2} \log(2\pi) - \frac{1}{2} (x^t - Ax^{t-1})^\top (x^t - Ax^{t-1}) \right].$$

Dropping the constant term  $-\frac{nT}{2} \log(2\pi)$  (which does not depend on  $A$ ), the log-likelihood as a function of  $A$  is:

$$\ell(A) = -\frac{1}{2} \sum_{t=1}^T (x^t - Ax^{t-1})^\top (x^t - Ax^{t-1}).$$

### Deriving the score equation and $\hat{A}_{\text{MLE}}$

We need to differentiate  $\ell(A)$  with respect to  $A$ . Using the trace identity from the hint, for a vector  $u \in \mathbb{R}^n$ :

$$u^\top u = \text{tr}(u^\top u) = \text{tr}(u u^\top).$$

Let  $u^t = x^t - Ax^{t-1}$ . Then:

$$\ell(A) = -\frac{1}{2} \sum_{t=1}^T \text{tr}((x^t - Ax^{t-1})(x^t - Ax^{t-1})^\top).$$

Expanding the product inside the trace:

$$(x^t - Ax^{t-1})(x^t - Ax^{t-1})^\top = x^t x^{t\top} - Ax^{t-1} x^{t\top} - x^t x^{t-1\top} A^\top + Ax^{t-1} x^{t-1\top} A^\top.$$

To compute the matrix derivative  $\frac{\partial \ell}{\partial A}$ , we use the standard matrix calculus identities:

- $\frac{\partial}{\partial A} \text{tr}(AB) = B^\top$
- $\frac{\partial}{\partial A} \text{tr}(ABA^\top) = A(B + B^\top)$

Therefore:

$$\begin{aligned}
\frac{\partial \ell}{\partial A} &= -\frac{1}{2} \sum_{t=1}^T \left[ -x^t x^{t-1\top} - x^t x^{t-1\top} + A \left( x^{t-1} x^{t-1\top} + x^{t-1} x^{t-1\top} \right) \right] \\
&= -\frac{1}{2} \sum_{t=1}^T \left[ -2 x^t x^{t-1\top} + 2 A x^{t-1} x^{t-1\top} \right] \\
&= \sum_{t=1}^T \left[ x^t x^{t-1\top} - A x^{t-1} x^{t-1\top} \right].
\end{aligned}$$

**Score equation.** Setting the score  $\frac{\partial \ell}{\partial A} = 0$ :

$$\sum_{t=1}^T x^t x^{t-1\top} = A \sum_{t=1}^T x^{t-1} x^{t-1\top}.$$

**MLE.** Solving for  $A$ :

$$\hat{A}_{\text{MLE}} = \left( \sum_{t=1}^T x^t x^{t-1\top} \right) \left( \sum_{t=1}^T x^{t-1} x^{t-1\top} \right)^{-1}.$$

This is the matrix analogue of the ordinary least squares estimator: we regress the “response”  $x^t$  on the “predictor”  $x^{t-1}$ , and the MLE is  $\hat{A} = S_{xy} S_{xx}^{-1}$ , where  $S_{xy} = \sum_t x^t x^{t-1\top}$  is the cross-covariance matrix and  $S_{xx} = \sum_t x^{t-1} x^{t-1\top}$  is the auto-covariance matrix of the lagged observations.

## Problem 28: Modeling overdispersion in RNA-seq gene expression (8 points)

### Part 1 (1 point)

We are given the hierarchical model:

$$Y_i \mid Z_i = z \sim \text{Pois}(z), \quad Z_i \sim \text{Gamma}(\alpha_i, \nu_i),$$

with  $\mathbb{E}(Z_i) = \mu_i = \alpha_i \nu_i$  and  $\text{Var}(Z_i) = \alpha_i^2 \nu_i$ .

#### Marginal mean

By the Law of Total Expectation:

$$\mathbb{E}(Y_i) = \mathbb{E}[\mathbb{E}(Y_i \mid Z_i)].$$

Since  $Y_i \mid Z_i \sim \text{Pois}(Z_i)$ , we have  $\mathbb{E}(Y_i \mid Z_i) = Z_i$ . Therefore:

$$\mathbb{E}(Y_i) = \mathbb{E}(Z_i) = \mu_i = \alpha_i \nu_i.$$

#### Marginal variance

By the Law of Total Variance:

$$\text{Var}(Y_i) = \mathbb{E}[\text{Var}(Y_i \mid Z_i)] + \text{Var}(\mathbb{E}(Y_i \mid Z_i)).$$

For a Poisson random variable,  $\text{Var}(Y_i \mid Z_i) = Z_i$  and  $\mathbb{E}(Y_i \mid Z_i) = Z_i$ . Substituting:

$$\text{Var}(Y_i) = \mathbb{E}(Z_i) + \text{Var}(Z_i) = \alpha_i \nu_i + \alpha_i^2 \nu_i.$$

$$\text{Var}(Y_i) = \alpha_i \nu_i + \alpha_i^2 \nu_i.$$

Since  $\alpha_i^2 \nu_i > 0$ , we have  $\text{Var}(Y_i) = \alpha_i \nu_i + \alpha_i^2 \nu_i > \alpha_i \nu_i = \mathbb{E}(Y_i)$ , confirming overdispersion.

### Interpretation of the two terms

- $\alpha_i \nu_i = \mathbb{E}(Z_i)$  corresponds to technical noise. This term equals  $\mathbb{E}[\text{Var}(Y_i | Z_i)]$ , i.e., the average Poisson variability given a fixed expression rate. It captures the stochasticity inherent in the sequencing process (amplification, reverse transcription, library preparation), which is modeled by the Poisson layer.
- $\alpha_i^2 \nu_i = \text{Var}(Z_i)$  corresponds to biological variability. This term equals  $\text{Var}(\mathbb{E}[Y_i | Z_i])$ , i.e., the variance of the latent expression rate across samples. It reflects the sample-to-sample differences in true gene expression levels, which inflate the total variance beyond what a pure Poisson model would predict.

### Part 2 (1 point)

**Model A** assumes a shared known shape parameter  $\nu_i = \nu$  for all samples. Since  $\alpha_i \nu_i = \mu_i$ , we have  $\alpha_i = \mu_i / \nu$ .

By the hint, marginalizing over  $Z_i$  yields the negative binomial distribution:

$$P(Y_i = y) = \binom{y + \nu - 1}{y} \left( \frac{\mu_i}{\nu + \mu_i} \right)^y \left( \frac{\nu}{\nu + \mu_i} \right)^\nu.$$

Taking the logarithm:

$$\log P(Y_i = y) = \log \binom{y + \nu - 1}{y} + y \log \left( \frac{\mu_i}{\nu + \mu_i} \right) + \nu \log \left( \frac{\nu}{\nu + \mu_i} \right).$$

### Identifying the EDF form

We define the canonical parameter:

$$\theta = \log \left( \frac{\mu_i}{\nu + \mu_i} \right).$$

From this,  $e^\theta = \frac{\mu_i}{\nu + \mu_i}$  and  $1 - e^\theta = \frac{\nu}{\nu + \mu_i}$ . Solving for  $\mu_i$ :

$$\mu_i = \frac{\nu e^\theta}{1 - e^\theta}.$$

Substituting into the log-PMF:

$$\log P(Y_i = y) = y\theta + \nu \log(1 - e^\theta) + \log \binom{y + \nu - 1}{y}.$$

This can be written in the EDF form  $\frac{y\theta - b(\theta)}{\phi} + c(y, \phi)$  with:

$$\phi = 1, \quad b(\theta) = -\nu \log(1 - e^\theta), \quad c(y, \phi) = \log \binom{y + \nu - 1}{y}.$$

Note that  $c(y, \phi)$  depends only on  $y$  and  $\nu$  (which is a known constant, not a function of  $\mu_i$ ), so it is independent of the mean parameter, confirming the EDF form.

### Deriving the variance function

The mean is:

$$\mu = b'(\theta) = -\nu \cdot \frac{-e^\theta}{1 - e^\theta} = \frac{\nu e^\theta}{1 - e^\theta}. \quad \checkmark$$

The variance function is:

$$V(\mu) = b''(\theta) = \nu \cdot \frac{d}{d\theta} \left( \frac{e^\theta}{1 - e^\theta} \right) = \nu \cdot \frac{e^\theta}{(1 - e^\theta)^2}.$$

Substituting  $e^\theta = \frac{\mu}{\nu + \mu}$  and  $1 - e^\theta = \frac{\nu}{\nu + \mu}$ :

$$V(\mu) = \nu \cdot \frac{\mu/(\nu + \mu)}{\nu^2/(\nu + \mu)^2} = \nu \cdot \frac{\mu(\nu + \mu)}{\nu^2} = \frac{\mu(\nu + \mu)}{\nu}.$$

$$V(\mu) = \mu + \frac{\mu^2}{\nu},$$

which is quadratic in  $\mu$ . The total variance is  $\text{Var}(Y_i) = \phi \cdot V(\mu) = \mu + \mu^2/\nu$ , consistent with Part 1 under  $\nu_i = \nu$  and  $\alpha_i = \mu_i/\nu$ .

### Part 3 (3 points)

**Model B** assumes a shared known scale parameter  $\alpha_i = \alpha$  for all samples. Since  $\alpha_i \nu_i = \mu_i$ , we have  $\nu_i = \mu_i/\alpha$ .

**Showing**  $V(\mu) = \mu$  **with**  $\phi = 1 + \alpha$

From Part 1, the marginal variance is:

$$\text{Var}(Y_i) = \alpha_i \nu_i + \alpha_i^2 \nu_i.$$

Substituting  $\alpha_i = \alpha$  and  $\nu_i = \mu_i/\alpha$ :

$$\text{Var}(Y_i) = \alpha \cdot \frac{\mu_i}{\alpha} + \alpha^2 \cdot \frac{\mu_i}{\alpha} = \mu_i + \alpha \mu_i = (1 + \alpha) \mu_i.$$

Writing this in the form  $\text{Var}(Y_i) = \phi \cdot V(\mu)$ , we identify:

$$V(\mu) = \mu, \quad \phi = 1 + \alpha,$$

so the total variance  $\text{Var}(Y_i) = (1 + \alpha)\mu$  is linear in  $\mu$ .

**Showing  $Y_i$  does not belong to the EDF**

Under Model B, the negative binomial PMF with  $\alpha_i = \alpha$  and  $\nu_i = \mu_i/\alpha$  is:

$$P(Y_i = y) = \binom{y + \mu_i/\alpha - 1}{y} \left( \frac{\mu_i}{\mu_i/\alpha + \mu_i} \right)^y \left( \frac{\mu_i/\alpha}{\mu_i/\alpha + \mu_i} \right)^{\mu_i/\alpha}.$$

Simplifying the probability terms using  $\mu_i/\alpha + \mu_i = \mu_i(1 + \alpha)/\alpha$ :

$$\frac{\mu_i}{\mu_i(1 + \alpha)/\alpha} = \frac{\alpha}{1 + \alpha}, \quad \frac{\mu_i/\alpha}{\mu_i(1 + \alpha)/\alpha} = \frac{1}{1 + \alpha}.$$

Therefore:

$$\log P(Y_i = y) = \log \binom{y + \mu_i/\alpha - 1}{y} + y \log \left( \frac{\alpha}{1 + \alpha} \right) + \frac{\mu_i}{\alpha} \log \left( \frac{1}{1 + \alpha} \right).$$

For this to be in the EDF form  $\frac{y\theta - b(\theta)}{\phi} + c(y, \phi)$ , the function  $c(y, \phi)$  must be independent of  $\mu_i$  (and hence of  $\theta$ ). However, the binomial coefficient  $\binom{y + \mu_i/\alpha - 1}{y}$  depends on both  $y$  and  $\mu_i$ . Expanding via the Gamma function:

$$\log \binom{y + \mu_i/\alpha - 1}{y} = \log \Gamma \left( y + \frac{\mu_i}{\alpha} \right) - \log \Gamma(y + 1) - \log \Gamma \left( \frac{\mu_i}{\alpha} \right).$$

The term  $\log \Gamma(y + \mu_i/\alpha)$  couples  $y$  and  $\mu_i$  in a nonlinear way that cannot be decomposed as a sum of a function of  $y$  alone and a function of  $\mu_i$  alone. Therefore, there is no valid decomposition into  $\frac{y\theta - b(\theta)}{\phi} + c(y, \phi)$  with  $c$  independent of  $\mu$ , and the distribution does not belong to the exponential dispersion family.

Note also that the second term  $y \log(\alpha/(1 + \alpha))$  is a constant (independent of  $\mu_i$ ), meaning the canonical parameter  $\theta$  would be constant — further confirming that the distribution cannot be parametrized in EDF form with  $\mu$  as the mean.

### Quasi-likelihood

Using the formula  $Q_i(\mu_i; y_i) = \int_{y_i}^{\mu_i} \frac{y_i - t}{\phi \cdot V(t)} dt$  with  $V(t) = t$  and  $\phi = 1 + \alpha$ :

$$Q_i(\mu_i; y_i) = \frac{1}{1 + \alpha} \int_{y_i}^{\mu_i} \frac{y_i - t}{t} dt = \frac{1}{1 + \alpha} \int_{y_i}^{\mu_i} \left( \frac{y_i}{t} - 1 \right) dt.$$

Evaluating:

$$Q_i(\mu_i; y_i) = \frac{1}{1 + \alpha} \left[ y_i \log t - t \right]_{y_i}^{\mu_i} = \frac{1}{1 + \alpha} \left[ y_i \log \mu_i - \mu_i - y_i \log y_i + y_i \right].$$

$$Q_i(\mu_i; y_i) = \frac{1}{1 + \alpha} \left[ y_i \log \left( \frac{\mu_i}{y_i} \right) - (\mu_i - y_i) \right].$$

This has the same functional form as the Poisson log-likelihood (up to the scaling factor  $1/(1 + \alpha)$ ), which is expected since the variance function  $V(\mu) = \mu$  is the same as for the Poisson distribution.

### Part 4 (3 points)

Now both parameters are allowed to vary with  $\mu_i$ :

$$\alpha_i = \gamma + \psi \mu_i, \quad \nu_i^{-1} = \psi + \gamma \mu_i^{-1}.$$

**Consistency check:**  $\alpha_i \nu_i = (\gamma + \psi \mu_i) \cdot \frac{1}{\psi + \gamma/\mu_i} = (\gamma + \psi \mu_i) \cdot \frac{\mu_i}{\psi \mu_i + \gamma} = \mu_i$ . ✓

**Special cases:** When  $\psi = 0$ :  $\alpha_i = \gamma$  (constant) and  $\nu_i = \mu_i/\gamma$ , recovering Model B. When  $\gamma = 0$ :  $\alpha_i = \psi \mu_i$  and  $\nu_i = 1/\psi$  (constant), recovering Model A.

### Deriving the variance function

From Part 1,  $\text{Var}(Y_i) = \alpha_i \nu_i + \alpha_i^2 \nu_i = \mu_i + \alpha_i^2 \nu_i$ .

We compute  $\alpha_i^2 \nu_i$ . Since  $\nu_i = \mu_i / \alpha_i = \mu_i / (\gamma + \psi \mu_i)$ :

$$\alpha_i^2 \nu_i = (\gamma + \psi \mu_i)^2 \cdot \frac{\mu_i}{\gamma + \psi \mu_i} = (\gamma + \psi \mu_i) \mu_i = \gamma \mu_i + \psi \mu_i^2.$$

Therefore:

$$\text{Var}(Y_i) = \mu_i + \gamma \mu_i + \psi \mu_i^2.$$

With  $\phi = 1$ , the variance function is:

$$V(\mu) = \mu + \gamma \mu + \psi \mu^2.$$

This interpolates between the two special cases:

- $\psi = 0$ :  $V(\mu) = (1 + \gamma)\mu$ , the overdispersed Poisson form (Model B with  $\alpha = \gamma$ ).
- $\gamma = 0$ :  $V(\mu) = \mu + \psi \mu^2 = \mu + \mu^2 / \nu$ , the negative binomial form (Model A with  $\nu = 1/\psi$ ).

### Showing $Y_i$ does not have the EDF form

The negative binomial PMF with  $\nu_i = \mu_i / (\gamma + \psi \mu_i)$  is:

$$P(Y_i = y) = \binom{y + \nu_i - 1}{y} \left( \frac{\gamma + \psi \mu_i}{1 + \gamma + \psi \mu_i} \right)^y \left( \frac{1}{1 + \gamma + \psi \mu_i} \right)^{\nu_i},$$

where we used  $\mu_i / (\nu_i + \mu_i) = (\gamma + \psi \mu_i) / (1 + \gamma + \psi \mu_i)$  and  $\nu_i / (\nu_i + \mu_i) = 1 / (1 + \gamma + \psi \mu_i)$ .

The log-PMF is:

$$\log P(Y_i = y) = \log \binom{y + \nu_i - 1}{y} + y \log \left( \frac{\gamma + \psi \mu_i}{1 + \gamma + \psi \mu_i} \right) + \nu_i \log \left( \frac{1}{1 + \gamma + \psi \mu_i} \right).$$

Since  $\nu_i = \mu_i / (\gamma + \psi \mu_i)$  depends on  $\mu_i$ , the binomial coefficient  $\binom{y + \nu_i - 1}{y}$  contains  $\log \Gamma(y + \nu_i)$ , which couples  $y$  and  $\mu_i$  in a way that cannot be decomposed into  $\frac{y\theta - b(\theta)}{\phi} + c(y, \phi)$  with  $c$  independent of  $\mu$ . Therefore, the distribution does not belong to the exponential dispersion family.

### Quasi-likelihood

Using  $Q_i(\mu_i; y_i) = \int_{y_i}^{\mu_i} \frac{y_i - t}{V(t)} dt$  with  $V(t) = t(1 + \gamma + \psi t)$ :

$$Q_i(\mu_i; y_i) = \int_{y_i}^{\mu_i} \frac{y_i - t}{t(1 + \gamma + \psi t)} dt.$$

We perform a partial fraction decomposition. Writing:

$$\frac{y_i - t}{t(1 + \gamma + \psi t)} = \frac{A}{t} + \frac{B}{1 + \gamma + \psi t},$$

we require  $y_i - t = A(1 + \gamma + \psi t) + Bt$ . Setting  $t = 0$ :  $A = \frac{y_i}{1 + \gamma}$ . Setting  $1 + \gamma + \psi t = 0$ :  $B = -\frac{1 + \gamma + \psi y_i}{1 + \gamma}$ .

**Verification:**  $A(1 + \gamma) + (A\psi + B)t = y_i + \left(\frac{y_i\psi}{1 + \gamma} - \frac{1 + \gamma + \psi y_i}{1 + \gamma}\right)t = y_i - t$ . ✓

Substituting and integrating:

$$Q_i = \frac{y_i}{1 + \gamma} \int_{y_i}^{\mu_i} \frac{dt}{t} - \frac{1 + \gamma + \psi y_i}{1 + \gamma} \int_{y_i}^{\mu_i} \frac{dt}{1 + \gamma + \psi t}.$$

The first integral is  $\log(\mu_i/y_i)$ . For the second, substituting  $u = 1 + \gamma + \psi t$ :

$$\int_{y_i}^{\mu_i} \frac{dt}{1 + \gamma + \psi t} = \frac{1}{\psi} \log\left(\frac{1 + \gamma + \psi \mu_i}{1 + \gamma + \psi y_i}\right).$$

Therefore:

$$Q_i(\mu_i; y_i) = \frac{y_i}{1 + \gamma} \log\left(\frac{\mu_i}{y_i}\right) - \frac{1 + \gamma + \psi y_i}{\psi(1 + \gamma)} \log\left(\frac{1 + \gamma + \psi \mu_i}{1 + \gamma + \psi y_i}\right).$$

### Verification of special cases:

- $\psi \rightarrow 0$  (**overdispersed Poisson, Model B**): Using  $\log(1 + x) \approx x$  for small  $x$ , the second term approaches  $\frac{1 + \gamma}{\psi(1 + \gamma)} \cdot \frac{\psi(\mu_i - y_i)}{1 + \gamma} = \frac{\mu_i - y_i}{1 + \gamma}$ , giving  $Q_i = \frac{1}{1 + \gamma} [y_i \log(\mu_i/y_i) - (\mu_i - y_i)]$ , consistent with Part 3.
- $\gamma = 0$  (**negative binomial, Model A**):  $Q_i = y_i \log(\mu_i/y_i) - \frac{1 + \psi y_i}{\psi} \log\left(\frac{1 + \psi \mu_i}{1 + \psi y_i}\right)$ , the negative binomial quasi-likelihood with  $\nu = 1/\psi$ .